॥वसुधैव कुटुम्बकम्॥

SYMBIOSIS
INSTITUTE OF TECHNOLOGY, PUNE

Project-based Learning I

Project Title: Vedology

By

1) PRN: 2407012262	Name: Abdul Wahid
2) PRN: 2407012280	Name: Mukund Prajapati
3) PRN: 2407012272	Name: Aseem Kelkar
4) PRN: 25070122513	Name: Het Bhalani

Under the Guidance of
Anuradha Pillai

SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

2024-25

Introduction

Vedology is a domain-specific Artificial Intelligence system designed for structured knowledge representation, retrieval, and contextual interpretation of classical Indian scriptures, including the Vedas and Upanishads. The system applies modern NLP techniques to traditionally unstructured textual corpora, enabling accurate semantic understanding while preserving original philosophical intent. Vedology functions as a scholarly interface between ancient knowledge systems and contemporary computational intelligence.

Motivation

Despite the historical, philosophical, and ethical significance of Indian scriptures, their academic adoption remains limited due to linguistic complexity, fragmented resources, and lack of intelligent interpretive tools. Existing digital resources focus primarily on digitization rather than understanding. Vedology is motivated by the need for an authoritative, AI-assisted system that enables systematic study, contextual analysis, and academic exploration of these texts without compromising textual integrity.

Statement of Problem

- Scriptural texts are inherently unstructured and semantically dense, making computational interpretation challenging.
- Current platforms provide translations but lack contextual reasoning, cross-textual correlation, and scholarly depth.
- Generic AI systems are not trained on Indic philosophical frameworks, leading to inaccuracies and oversimplifications.

Objectives of Study

- To develop a domain-trained AI system capable of precise semantic understanding of Indian scriptures.
- To implement structured knowledge representation for complex philosophical concepts.
- To ensure interpretive accuracy through controlled datasets and validated sources.
- To facilitate academic research and structured learning.
- To demonstrate the application of AI in preserving and analyzing non-Western knowledge systems.

Existing Systems

	Traditional Commentaries	Digital Libraries and Archives	General-Purpose AI Systems
Authority	Highly authoritative	No inherent authority	Low authority
Accessibility	Inaccessible	Accessible	Accessible
Interpretation	High	Low	Low
Specialization	High	Low	Low

Comparison of Existing Systems

	Traditional Sources	Digital Archives	Generic AI Systems	Vedology
Domain Specialization	High	Medium	Low	High
Contextual Accuracy	High	Low	Low	High
Structured Knowledge	Low	Medium	Low	High
Interpretive Depth	High	Low	Low	High
Academic Reliability	High	Medium	Low	High

Conclusion

Vedology demonstrates how Artificial Intelligence can be responsibly applied to classical knowledge systems without compromising scholarly integrity. By transforming unstructured philosophical texts into an intelligent, interpretable, and academically reliable system, Vedology establishes a framework for future research at the intersection of AI, humanities, and cultural preservation. The project validates AI's role beyond automation—towards intellectual augmentation and knowledge preservation.

References

- Rig Veda, Atharva Veda, Yajur Veda, Sama Veda
- Principal Upanishads (Isha, Katha, Chandogya, Brihadaranyaka)
- Adi Shankaracharya – Classical Commentaries
- Research Literature on Natural Language Processing and Knowledge Representation
- Digital Sanskrit Text Repositories and Academic Archives